



Exploring Big Data on a Desktop: Elasticsearch on OpenStack

This column explores the use of the powerful, open source, distributed search tool, Elasticsearch, which can enable the retrieval of data through a simple search interface.

When you have a huge number of documents, wouldn't it be great if you could search them almost as well as you can with Google? Lucene (<http://lucene.apache.org/>) has been helping organisations search their data for years. Projects like Elasticsearch (<http://www.elasticsearch.org/>) are built on top of Lucene to provide distributed, scalable solutions in order to search huge volumes of data. A good example is the use of Elasticsearch at WordPress (<http://gibrown.wordpress.com/2014/01/09/scaling-elasticsearch-part-1-overview/>).

In this experiment, you start with three nodes on OpenStack: *h-mstr*, *h-slv1* and *h-slv2* as in the previous article. Download the rpm package from the Elasticsearch site and install it on each of the nodes.

The configuration file is */etc/elasticsearch/elasticsearch.yml*. You will need to configure it on each of the three nodes. Consider the following settings on the *h-mstr* node:

```
cluster.name: es
node.master: true
node.data: true
index.number_of_shards: 10
index.number_of_replicas: 0
```

We have given the name *es* to the cluster. The same value should be used on the *h-slv1* and *h-slv2* nodes. The *h-mstr* node will act as a master and store data as well. The master nodes process the requests by distributing the search to the data nodes and consolidating the results. The next two parameters relate to the index. The number of shards is the number of sub-indices that are created and distributed among the data nodes. The default value for the number of shards is 5. The number of replicas represents the additional copies of the indices created. Since it has been set to *No replicas*, the default value is 1.

You may use the same values on *slv1* and *slv2* nodes or use *node.master* set to *False*. Once you have loaded the data, you will find that the *h-mstr* node has four shards and *h-slv1* and *h-slv2* have three shards each. The indices will be in the directory */var/lib/elasticsearch/es/nodes/0/indices/* on each node.

You start Elasticsearch on each node by executing the following command:

```
$ sudo systemctl start elasticsearch
```

You can get to know the status of the cluster by browsing http://h-mstr:9200/_cluster/health?pretty.

Loading the data

If you want to index the documents located on your desktop, Elasticsearch supports a Python interface for it. It is available in the Fedora 20 repository. So, on your desktop, install:

```
$ sudo yum install python-elasticsearch
```

The following is a sample program to index LibreOffice documents. The comments embedded in the code hopefully make it clear that this is not a complex task.

```
#!/usr/bin/python
import sys
import os
import subprocess
from elasticsearch import Elasticsearch

FILETYPES=['odt','doc','sxw','abw']
# Convert a document file into a text file in /tmp and return
the text file name
def convert_to_text(inpath,infile):
    subprocess.call(['soffice','--headless','--convert-
to','txt:Text',
        '--outdir','/tmp','/'.join([inpath,infile])])
    return '/tmp/' + infile.rsplit('.',1)[0] + '.txt'
# Read the text title and return it as a string
def process_file(p,f):
    textfile = convert_to_text(p,f)
    return ' '.join([line.strip() for line in open(textfile)])
# Search all files in a root path and select the document files
def get_documents(path):
    for curr_path,dirs,files in os.walk(path):
        for f in files:
            try:
                if f.rsplit('.',1)[1].lower() in FILETYPES:
                    yield curr_path,f
            except:
                pass
# Run this program with the root directory.
```